

Postpartum Depression And It's Impact Analysis

Md. Tahiadur Rahman (ID: 20200204003)*

Adiba Amin (ID: 20200204012)*

Md. Rafiu Alam Rafi (ID: 20200204051)*

Antika Ghosh (ID: 190104005)*

*Department of CSE, Ahsanullah University of Science and Technology, Dhaka, Bangladesh

Email: {rahman.cse.20200204003, adiba.cse.20200204012, rafi.cse.20200204051, 190104005}@aust.edu

Abstract—This study investigates the prediction of postpartum depression (PPD) through the application of various machine learning (ML) models, analyzing a comprehensive dataset from a cohort of 4,313 postpartum women. Key predictive features identified include irritability towards baby and partner, trouble sleeping at night, feeling sad or tearful, feeling of guilt, overeating or loss of appetite, and problems concentrating or making decisions. The study implemented a diverse range of models, such as Decision Trees, Random Forests, Gradient Boosting, K-Nearest Neighbors (KNN), Support Vector Machines (SVM), Neural Networks, and Naïve Bayes classifiers. Among these, the Decision Tree and Random Forest models demonstrated the highest accuracies of 92% and 91%, respectively, while Gradient Boosting, KNN, SVM, and Neural Networks showed accuracies between 74% and 83%. The Naïve Bayes classifier had the lowest accuracy at 60%. This research highlights the robustness of tree-based models in predicting PPD and emphasizes the potential for these models to be integrated into healthcare systems for early identification and intervention strategies, ultimately improving maternal mental health outcomes.

Index Terms—postpartum depression, machine learning, feature selection, tree-based models, mental health, predictive analytics, SVM (Support Vector Machine), Suicide Attempt, Clinical Prediction, Naïve Bayes, Public Health

I. INTRODUCTION

Postpartum depression (PPD) represents a significant mental health challenge affecting a substantial number of new mothers worldwide, with profound implications for maternal well-being, infant care, and family dynamics. While the condition's prevalence underscores its importance, diagnosing PPD accurately remains a complex endeavor due to its multifaceted nature and varied presentation among individuals. Traditional diagnostic approaches often rely on subjective assessments and clinical judgment, which can lead to inconsistencies and delays in intervention. Recognizing this gap, recent advancements in machine learning (ML) offer a promising avenue to enhance PPD detection through objective analysis of diverse symptom profiles and risk factors. By leveraging computational algorithms, this study aims to harness the power of ML to develop robust predictive models capable of identifying PPD with high accuracy and efficiency.

The research investigates a comprehensive set of predictive features encompassing emotional indicators such as irritability towards the baby and partner, disturbances in sleep patterns, feelings of sadness or guilt, changes in appetite, and difficulties in concentration. Through the application of various ML

techniques—including Decision Trees, Random Forests, Gradient Boosting, K-Nearest Neighbors (KNN), Support Vector Machines (SVM), Neural Networks, and Naïve Bayes classifiers—the study evaluates their effectiveness in discerning patterns within these complex datasets. Key aspects of the study include the interpretability of Decision Trees and Random Forests in uncovering nuanced relationships among symptoms, as well as advanced techniques like feature selection to enhance model performance. By systematically evaluating these ML models, the research seeks to establish reliable tools for early PPD detection, thereby facilitating timely intervention and personalized care strategies. Ultimately, integrating ML-driven predictive models into clinical practice holds the potential to transform PPD diagnosis, enabling healthcare professionals to deliver targeted interventions that improve outcomes for postpartum women and their families. This interdisciplinary approach underscores the intersection of technology and healthcare in addressing mental health challenges, paving the way for more effective and compassionate maternal care practices globally.

II. MOTIVATION

The increasing awareness of postpartum depression (PPD) highlights the urgent need to understand its severe consequences, including the risk of suicide. While mental health research has advanced significantly, studies specifically targeting PPD-related suicide remain limited. This gap is critical given the profound impact PPD has on mothers and families. By analyzing numeric data to predict suicide risk, our research aims to provide healthcare professionals with actionable insights. These insights can lead to more effective intervention strategies and support systems, ultimately saving lives and improving mental health outcomes. Addressing this issue not only enhances patient care but also contributes to the broader effort of integrating data-driven approaches into mental health services. As the prevalence of PPD continues to rise, developing predictive tools becomes increasingly essential for timely and effective intervention. This study aims to fill a vital gap and support the broader goal of improving mental health care for new mothers.

III. LITERATURE REVIEW

The integration of machine learning (ML) techniques for predicting postpartum depression (PPD) marks a substantial

advancement in maternal mental health research. Traditionally, diagnosing PPD relied on subjective assessments and clinical interviews, potentially missing crucial symptoms and risks. By employing ML algorithms, researchers enhance predictive accuracy by analyzing diverse datasets containing demographic, psychological, and behavioral factors. These methods aim to uncover intricate patterns and predictors of PPD, enabling earlier detection and intervention strategies. These researches highlights the evolving applications of ML in PPD studies, emphasizing methodologies, findings, and implications for improving maternal mental health outcomes through data-driven approaches.

In this paper [1], machine learning (ML) techniques have been shown to predict postpartum depression (PPD) using large datasets. A scoping review identified 14 studies employing supervised learning methods such as support vector machine (SVM), random forest, Naive Bayes, regression, artificial neural network, decision trees, and XGBoost. The studies reported varied performance, with SVM achieving an AUC of 0.78-0.86, random forest 0.88, XGBoost 0.80, and logistic regression 0.93. These results suggest that ML can significantly improve early PPD detection, though further clinical research is needed to optimize these algorithms for clinical practice.

This paper [2] states that postpartum depression (PPD) poses a significant health concern, and machine learning (ML) has emerged as a promising tool for predicting its risk. A systematic review encompassing 17 studies and 62 prediction models revealed that supervised learning, particularly using support vector machine, random forest, and logistic regression models, predominated in PPD prediction. Despite this, the studies generally exhibited high risk of bias and limited validation, with only a minority conducting both internal and external validation. Translation of models into clinical practice remains scarce, highlighting a gap between model development and clinical implementation. Further efforts are needed to enhance the quality and validation of ML models to effectively deploy predictive tools for PPD in clinical settings.

Postpartum depression, affecting 15% of US mothers, is associated with significant adverse outcomes. This study [3] developed predictive models using machine learning (ML) on PRAMS 2012–2013 data (28,755 records). Nine ML algorithms were employed, with random forest achieving the highest AUC of 0.884, followed by SVM at 0.864. The models addressed data imbalance and achieved classification accuracies ranging from 0.650 (kNN) to 0.791 (RF). Maternal demographic factors significantly associated with PPD included education, smoking status, marital status, and pre-pregnancy depression. These findings underscore ML's potential as a predictive tool for postpartum depression screening and intervention.

This study [4] aimed to predict postpartum depressive (PPD) symptoms using machine learning (ML) methods based on clinical, demographic, and psychometric data collected after childbirth. It employed various ML algorithms, including Ridge Regression, LASSO Regression, Gradient Boosting, and

others, evaluating their performance metrics such as accuracy, sensitivity, specificity, NPV, and AUC. Despite varying model performances, XRT emerged as the most robust with an accuracy of 72% and AUC of 79%. Key predictors of PPD included anxiety and depressive symptoms during pregnancy, along with factors related to resilience and personality. The study highlighted the potential of ML in identifying high-risk mothers at discharge for targeted preventive interventions, despite challenges like low PPV due to PPD's prevalence.

The study [5] developed a predictive framework for postpartum depression (PPD) using machine learning algorithms like Ridge Regression, Gradient Boosting, and Extreme Randomized Forests, optimized through Sequential Forward Selection and parameter tuning via grid search. XRT emerged as the most accurate model with 72% accuracy and 79% AUC, identifying predictors such as anxiety, mood disorders, antidepressant use, and pregnancy-related factors like cesarean section. The framework leverages electronic health record data to update risk assessments over time, potentially enhancing clinical interventions and screening practices for PPD.

This study [6] aimed to develop predictive models for postpartum depression (PPD) using pregnancy data, comparing SVM and RF algorithms with two feature selection methods. Analyzing 508 women, predictors included demographics, social factors, and mental health indicators, with Edinburgh Postnatal Depression Scale scores post-delivery as outcomes. SVM with FFS-RF showed superior prediction efficacy (sensitivity=0.69, AUC=0.78). RF highlighted key predictors: psychological resilience, third trimester depression, and income level. These findings emphasize early identification using maternal psychological resilience for effective PPD risk mitigation.

This study [7] states that Postpartum depression (PPD) affects up to 25% of mothers within the first year postpartum, with severe health risks for both mothers and infants. Traditional screening methods, such as the Edinburgh Postnatal Depression Scale, often miss about 50% of cases. Recent studies have explored machine learning (ML) techniques to improve PPD prediction. A review of 11 studies using ML, primarily support vector machines, found promising results with an area under the curve (AUC) above 0.7, indicating feasible prediction. These studies predominantly utilized sociodemographic and clinical variables, with few incorporating biological data. Despite limited research, ML approaches hold potential for enhancing PPD screening and effectively targeting at-risk populations.

This study [8] state that, Postpartum depression (PPD) affects 12% of new mothers, causing significant health impacts. Despite available treatments, many women remain undiagnosed due to poor identification methods. This study utilized clinical, demographic, and psychometric data from the BASIC cohort (n=4313) in Uppsala, Sweden, to evaluate machine learning (ML) methods for predicting PPD. The extremely randomized trees (XRT) method showed robust performance with 73% accuracy, 72% sensitivity, 75% specificity, 33% positive predictive value, 94% negative predictive value, and an area under the curve (AUC) of 81%. Key

predictors included depression and anxiety during pregnancy, resilience, and personality traits. Other ML models evaluated were Ridge Regression,

IV. DATASET AND METHODOLOGY

A. Dataset

In our research study, we gathered a dataset comprising 1503 records from a medical hospital using a structured questionnaire administered through a Google Form. This dataset, which remains unpublished, focuses on understanding postnatal depression (PDD) and related psychological states among participants. The dataset includes 15 attributes, from which we selected 10 for analysis, with "Feeling Anxious" designated as the target attribute. Attributes such as age, irritability towards baby and partner, problems concentrating or making decisions, and feelings of sadness or tearfulness were meticulously chosen to explore their impact on feelings of anxiety postnatally. Upon data collection, rigorous preprocessing steps were implemented to ensure data quality and relevance. This included handling missing data and encoding categorical variables into numerical values for compatibility with machine learning algorithms. Feature selection techniques, such as Recursive Feature Elimination (RFE), were employed to identify the most influential attributes affecting feelings of anxiety.

Statistical analysis and model building were pivotal aspects of our study. We utilized a range of machine learning algorithms including Logistic Regression, Random Forest, Naive Bayes, Gradient Boosting, K-Nearest Neighbors (KNN), and Neural Networks. Each algorithm was chosen for its unique strengths in predicting and understanding the occurrence of anxiety based on the selected attributes. Evaluation metrics such as accuracy, precision, recall, and F1-score were employed to assess the performance of these models, with confusion matrices providing visual insights into their predictive capabilities. Our findings provide valuable insights into the factors contributing to feelings of anxiety postnatally, shedding light on potential interventions and support strategies for affected individuals. This dataset serves as a foundational resource for future research aimed at enhancing mental health care in postnatal settings, emphasizing the importance of personalized approaches tailored to individual psychological states and experiences.

B. Feature Extraction

Feature selection in this study was a critical step aimed at identifying and prioritizing the most informative variables associated with postpartum depression (PPD). The process began with preprocessing the dataset to ensure all data were in a suitable format for analysis, including mapping categorical variables to numerical values for consistency across the dataset.

Several advanced techniques were employed to select features that were most predictive of PPD severity. Recursive Feature Elimination (RFE) was one of the primary methods used. RFE iteratively selected features by training a model and eliminating the least significant ones until the desired number of

features was reached. In this study, RFE identified and retained six key features: Feeling sad or Tearful, Irritability towards baby and partner, Trouble sleeping at night, Overeating or loss of appetite, Feeling anxious, and Feeling of guilt. These features were chosen based on their ability to best differentiate between different levels of PPD severity, as determined by their coefficients in the logistic regression model.

Additionally, SelectKBest with chi-squared scoring was employed to rank all features based on their statistical significance relative to the target variable (PPD severity). This statistical test evaluated the independence between each feature and the target, identifying those with the strongest relationships. By selecting the top features according to their chi-squared scores, the study ensured that only the most relevant variables contributed to the predictive models.

Furthermore, ensemble learning methods such as Extra Trees Classifier and Random Forests were utilized to assess feature importance based on their ability to decrease impurity in classification tasks. These models evaluated each feature's contribution to predictive accuracy, thereby highlighting those that provided the most discriminatory power in identifying PPD.

In summary, the feature selection process combined sophisticated statistical techniques with machine learning methodologies to pinpoint the most critical symptoms and indicators of PPD. By prioritizing these features, the study not only enhanced the accuracy of predictive models but also provided actionable insights into the underlying factors contributing to maternal mental health challenges. This comprehensive approach underscores the importance of targeted feature selection in understanding and addressing postpartum depression effectively.

V. MODEL ARCHITECTURE

In our research focused on postnatal depression (PDD), we collected a dataset comprising 1503 records from a medical hospital via a questionnaire administered through Google Forms. This dataset includes 15 attributes, from which we selected 10 for detailed analysis. These attributes encompass various aspects related to postnatal mental health, such as emotional states, sleeping patterns, bonding with the baby, and feelings of anxiety. Our primary focus was on understanding and predicting the occurrence of "Feeling Anxious" as the target attribute among postnatal individuals. To ensure the dataset's suitability for analysis, rigorous preprocessing steps were undertaken. This involved cleaning the data to handle missing values and encoding categorical variables into numerical formats. Feature engineering techniques were applied to transform and select the most relevant attributes for our predictive models. Techniques like Recursive Feature Elimination (RFE) were employed to identify key features contributing to feelings of anxiety postnatally.

For model development, we implemented a range of machine learning algorithms tailored to classification tasks. These included Logistic Regression for linear classification, Random Forest for ensemble-based decision-making, Naive Bayes for

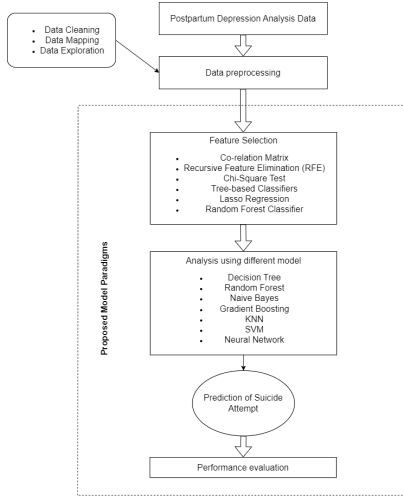


Fig. 1. Workflow of the overall process

probabilistic modeling, Gradient Boosting for sequential ensemble learning, K-Nearest Neighbors (KNN) for instance-based classification, and Neural Networks for complex pattern recognition. Each model was trained and evaluated using standard performance metrics such as accuracy, precision, recall, and F1-score, along with visual representations like confusion matrices to assess predictive capabilities. Our approach not only aimed to predict feelings of anxiety but also sought deeper insights into the factors contributing to postnatal mental health challenges. The architecture of our study emphasized a systematic workflow from data preprocessing through model training and evaluation, ultimately yielding insights crucial for developing targeted interventions and supporting individuals experiencing postnatal depression. Here, seven prevalent classifiers were utilized - Decision Trees, Random Forests, Naïve Bayes, Gradient Boosting, KNN, Support Vector Machine (SVM) and Neural Network.

Random Forests consist of an ensemble of decision trees, each trained on a randomized subset of the data features. By aggregating predictions across many such decision trees, the overall model becomes robust to noise and overfitting. Mathematically, for an input feature vector $\mathbf{x}$, each decision tree t outputs a class prediction $\hat{y}_t(\mathbf{x})$. The forest combines these predictions through a weighted voting scheme to obtain the overall prediction as:

$$\hat{y}(\mathbf{x}) = \sum_t w_t \hat{y}_t(\mathbf{x}) \quad (1)$$

where w_t denotes the weight applied to decision tree t 's prediction.

Decision Trees recursively partition the feature space into homogeneous splits based on cutting thresholds in individual features. This partitioning forms a tree structure where leaf nodes correspond to class outcomes. Mathematically, for a feature vector $\mathbf{x}$, the tree can be defined as a set of splitting rules $r_j(\mathbf{x})$, such that:

$$\hat{y}(\mathbf{x}) = \sum_j I(r_j(\mathbf{x}))c_j \quad (2)$$

where $I(\cdot)$ is an indicator function evaluating to 1 if $\mathbf{x}$ satisfies rule r_j , and c_j is the class label for leaf j .

Support Vector Machines (SVMs) find an optimal hyperplane to separate classes in the high-dimensional feature space. Maximizing the margin between classes enables even non-linearly separable data to be classified. Using a kernel function $\Phi(\mathbf{x})$, SVMs solve:

$$\min \|\mathbf{w}\|^2 \quad \text{s.t.} \quad y_i(\Phi(\mathbf{x}_i) \cdot \mathbf{w} + b) \geq 1 \quad (3)$$

Here, $\mathbf{w}$ denotes the hyperplane normal vector, b the intercept, and y_i the class labels. This produces a decision boundary given by:

$$f(\mathbf{x}) = \text{sign} \left(\sum_i \alpha_i y_i \Phi(\mathbf{x}_i) \cdot \Phi(\mathbf{x}) + b \right) \quad (4)$$

where α_i are optimized weights for support vectors $\mathbf{x}_i$.

Naïve Bayes classifiers are based on Bayes' theorem with strong (naïve) independence assumptions between the features. They are simple yet effective, particularly for text classification tasks. In this study, Gaussian Naïve Bayes was applied, which assumes that the features follow a normal distribution. The classification rule is given by:

$$\hat{y} = \arg \max_y P(y) \prod_{i=1}^n P(x_i | y) \quad (5)$$

where $P(y)$ is the prior probability of class y and $P(x_i | y)$ is the likelihood of feature x_i given class y .

Gradient Boosting builds an ensemble of trees sequentially, where each tree corrects the errors made by the previous one. It's a powerful technique for classification tasks, often providing improved accuracy compared to individual decision trees. The model is built by minimizing the following loss function:

$$F_{m+1}(\mathbf{x}) = F_m(\mathbf{x}) + \eta h_m(\mathbf{x}) \quad (6)$$

where $F_m(\mathbf{x})$ is the current model, $h_m(\mathbf{x})$ is the new tree, and η is the learning rate.

K-Nearest Neighbors (KNN) is a non-parametric method used for classification and regression tasks. It assigns a class label to an instance based on the majority class among its K nearest neighbors in the feature space. The predicted class $\hat{y}$ for an instance $\mathbf{x}$ is given by:

$$\hat{y} = \arg \max_y \sum_{i \in \mathcal{N}_K(\mathbf{x})} I(y_i = y) \quad (7)$$

where $\mathcal{N}_K(\mathbf{x})$ denotes the set of K nearest neighbors of $\mathbf{x}$ and $I(\cdot)$ is an indicator function.

Neural Networks (NN) consist of layers of interconnected neurons that can learn complex patterns. They are highly flexible and can capture intricate relationships between features.

In this study, a Multi-Layer Perceptron (MLP) classifier was used, which is a type of feedforward neural network. The output of an MLP with one hidden layer is given by:

$$\mathbf{h} = \sigma(\mathbf{W}_1 \mathbf{x} + \mathbf{b}_1) \quad (8)$$

$$\hat{y} = \sigma(\mathbf{W}_2 \mathbf{h} + \mathbf{b}_2) \quad (9)$$

where σ is an activation function, $\mathbf{W}_1$ and $\mathbf{W}_2$ are weight matrices, $\mathbf{b}_1$ and $\mathbf{b}_2$ are bias vectors, and $\mathbf{h}$ is the hidden layer output.

VI. RESULT ANALYSIS AND COMPARISON

Table 1 presents the performance of various models through a weighted average of three metrics: precision, recall, and F1-score. The analysis indicates that Decision Tree and Random Forest models offer balanced performance with high precision (0.94), recall (0.93 and 0.91), and accuracy (0.92 and 0.91). Conversely, the Naïve Bayes model shows limitations, with lower precision (0.62), recall (0.73), and accuracy (0.60), indicating issues with false positives and negatives.

TABLE I
PERFORMANCE METRICS OF DIFFERENT MODELS

Model	Precision	Recall	F1-Score	Accuracy
Decision Tree	0.94	0.93	0.94	0.92
Random Forest	0.94	0.91	0.93	0.91
Naïve Bayes	0.62	0.73	0.67	0.60
Gradient Boosting	0.85	0.90	0.88	0.83
KNN	0.82	0.91	0.86	0.82
SVM	0.78	0.90	0.83	0.75
Neural Network	0.80	0.89	0.84	0.74

The Gradient Boosting model performs well with precision (0.85), recall (0.90), and accuracy (0.83), while the KNN model also shows strong results (precision 0.82, recall 0.91, accuracy 0.82). The SVM model, despite good recall (0.90), has lower precision (0.78) and accuracy (0.75), suggesting false positive issues. The Neural Network model, with precision (0.80), recall (0.89), and accuracy (0.74), highlights the potential of deep learning but may need further tuning. This analysis underscores that simpler models can be effective, but complex data often requires sophisticated models for accurate analysis.

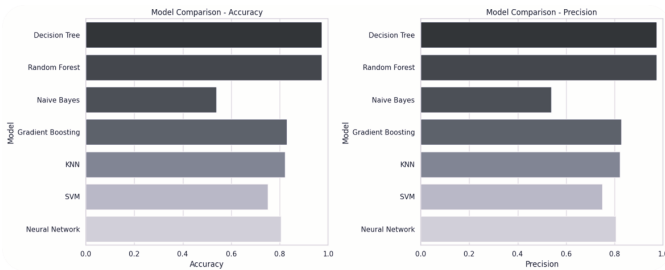


Fig. 2. Model Comparison: Accuracy and Precision

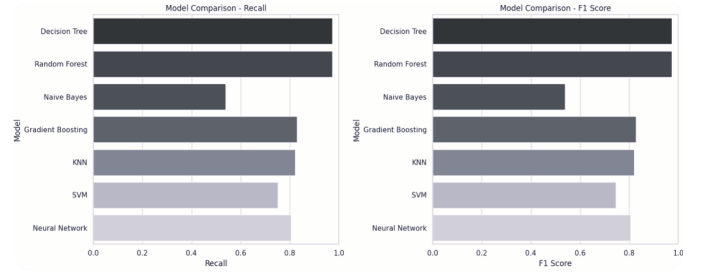


Fig. 3. Model Comparison: Recall and F1 Score

VII. DISCUSSION

This study [9] aimed to identify key predictors and evaluate the performance of various machine learning models in predicting postpartum depression (PPD), with a particular focus on the risk of suicide attempts among postpartum women. The process began with data visualization and feature mapping, which transformed categorical variables into numerical values, thereby simplifying the modeling process. By dropping missing values and irrelevant columns, we ensured that the dataset was clean and suitable for analysis.

Feature selection was a crucial step in this analysis. Using Recursive Feature Elimination (RFE), we identified six key features that significantly contributed to predicting suicide attempts: 'Feeling sad or Tearful,' 'Irritable towards baby and partner,' 'Trouble sleeping at night,' 'Overeating or loss of appetite,' 'Feeling anxious,' and 'Feeling of guilt.' These features were consistently highlighted by other selection methods, including SelectKBest, ExtraTreesClassifier, and LassoCV, indicating their robust importance across different statistical and machine learning approaches. This consistency underscores the clinical relevance of these emotional and behavioral indicators in assessing the risk of PPD and associated suicidal behaviors.

In comparing the performance of different classification models, the Decision Tree Classifier emerged as the most accurate, achieving a 92% accuracy rate. This model's performance can be attributed to its ability to handle complex interactions between features and its straightforward interpretability, which is valuable in clinical settings. The Random Forest Classifier followed closely with an accuracy of 91%, benefiting from its ensemble nature that combines multiple decision trees to improve generalizability and robustness.

The Gradient Boosting Classifier, with an accuracy of 83%, also demonstrated effective performance, particularly in balancing bias and variance through its iterative boosting approach. This suggests that ensemble methods, which aggregate the strengths of multiple learners, are particularly well-suited for this type of predictive task. In contrast, the Naive Bayes Classifier, despite its simplicity and computational efficiency, showed the lowest accuracy at 60%. This lower performance likely results from its assumption of feature independence, which does not hold true for the complex, interrelated symptoms of PPD.

Support Vector Machines (SVM) and K-Nearest Neighbors (KNN) classifiers showed moderate performance with accuracies of 75% and 82%, respectively. These models require careful tuning of parameters and are sensitive to the choice of distance metrics and kernel functions, which may account for their relative underperformance compared to tree-based models. The Neural Network classifier achieved an accuracy of 74%, similar to SVM. This suggests that while neural networks have the potential to model complex patterns, they require larger datasets and more extensive parameter tuning to reach their full potential.

The findings of this study highlight the complexity of predicting PPD and suicide attempts, emphasizing the need for comprehensive approaches in both feature selection and model application. The identified key features, including emotional indicators like sadness, irritability, and guilt, and behavioral symptoms such as sleep disturbances and overeating, align with clinical observations. This alignment reinforces their importance in screening and intervention strategies for PPD.

Practically, the high accuracy of tree-based models suggests their potential utility in clinical settings. These models could support healthcare providers in identifying high-risk individuals, enabling timely and tailored interventions. The interpretability of these models is particularly advantageous, allowing clinicians to understand the basis for predictions and make informed decisions. However, the study also highlights the necessity for further research. Integrating additional data sources, such as longitudinal tracking, genetic information, and more granular clinical data, could enhance the predictive capabilities of these models.

Additionally, addressing potential biases in the models and ensuring they are validated across diverse populations is crucial for broad applicability. This would involve rigorous cross-validation and possibly incorporating fairness metrics to ensure that predictions are equitable across different demographic groups.

VIII. LIMITATIONS AND FUTURE WORK

A. Limitations of the Study

The study [9] on predicting postnatal depression using machine learning models, based on a dataset collected from a single medical hospital via a Google form questionnaire, presents several noteworthy limitations.

Firstly, the dataset's reliance on a specific hospital and the mode of data collection through Google forms may introduce biases. These biases could stem from the demographics, socio-economic status, and health-seeking behavior of the participants within that hospital's catchment area. As a result, the findings may not generalize well to broader populations, highlighting the need for datasets collected from multiple sources to improve external validity.

Secondly, the selection of features crucially impacts the models' performance. While the study carefully chose attributes deemed relevant to postnatal depression, there's a possibility that important predictors were omitted. Future research could benefit from incorporating additional variables such

as social support networks, environmental factors, or genetic predispositions, which could potentially enhance predictive accuracy and broaden the understanding of the condition's multifactorial nature.

Moreover, despite achieving reasonable predictive accuracy, the study's models may not have reached their full potential. Further optimization of model hyperparameters, exploration of alternative algorithms (e.g., ensemble methods, deep learning architectures), and robust validation across diverse datasets are essential steps to maximize the models' efficacy and applicability in clinical settings.

Furthermore, the imbalance in the distribution of classes within the target variable ("Feeling Anxious") poses a challenge. This imbalance can skew model performance metrics, especially for minority classes, and necessitates the adoption of strategies like class weighting, data resampling techniques (e.g., SMOTE), or alternative evaluation metrics to mitigate potential biases and ensure fair and accurate model assessments.

Lastly, the interpretability of complex models, such as neural networks and ensemble methods used in the study, remains a concern. In clinical contexts, where transparency and understanding of model decisions are crucial, developing methods for model interpretation—such as feature importance analysis, model visualization, or generating explanations for predictions—becomes imperative to foster trust and facilitate informed decision-making.

B. Future Work

To address these limitations and advance the research on predicting postnatal depression, several promising avenues for future work can be explored.

Firstly, researchers could focus on prospective data collection efforts, including longitudinal studies or multi-center collaborations. By gathering data from diverse populations and settings, researchers can enhance the robustness and generalizability of predictive models, thereby improving their utility in real-world clinical practice.

Secondly, enhancing feature engineering techniques represents a pivotal step forward. Introducing novel features and leveraging advanced feature selection methodologies—such as recursive feature elimination (RFE), genetic algorithms, or domain-specific knowledge integration—can further enrich the predictive capabilities of models and uncover hidden relationships contributing to postnatal depression risk.

Additionally, exploring ensemble methods that combine the strengths of different modeling approaches (e.g., decision trees, logistic regression, support vector machines) could yield superior predictive performance. Techniques like model stacking, blending, or hybrid architectures may offer synergistic advantages, enhancing model robustness and reliability across diverse datasets.

Moreover, addressing the challenge of imbalanced data through innovative techniques—such as adaptive sampling methods, ensemble-based approaches, or hybrid resampling strategies—can foster equitable model predictions across all

classes of the target variable, thereby improving the models' practical utility and clinical relevance.

Furthermore, advancements in explainable artificial intelligence (AI) techniques are critical. Developing interpretable models and methodologies for model explanation—such as SHAP (Shapley Additive Explanations), LIME (Local Interpretable Model-agnostic Explanations), or feature importance visualization—can empower healthcare professionals to understand, validate, and trust the predictions made by complex machine learning models in clinical decision-making contexts.

Lastly, rigorous clinical validation and deployment studies are essential. Collaborating closely with healthcare providers and stakeholders to evaluate models in real-world clinical settings can validate their efficacy, feasibility, and ethical implications. Such validation studies are crucial for integrating predictive models into routine healthcare practices, ensuring they contribute positively to early detection, intervention, and personalized management of postnatal depression.

REFERENCES

- [1] Kiran Saqib, MBBS, MSPH; Amber Fozia Khan, BSc; Zahid Ahmad Butt, MBBS, MSc, PhD, 'Machine Learning Methods for Predicting Postpartum Depression: Scoping Review', 24.11.2021 in Vol 8, No 11 (2021): November
- [2] Minhui Zhong, Chan Yu, Jinxia Jiang, Xia Duan, 'Application of machine learning in predicting the risk of postpartum depression: A systematic review', Volume 318, 1 December 2022, Pages 364-379
- [3] Dayeon Shin, Kyung Ju Lee, Temidayo Adeluwa, and Junguk Hur, 'Machine Learning-Based Predictive Modeling of Postpartum Depression', 8 September 2020
- [4] Sam Andersson, Deepti R. Bathula, Stavros I. Iliadis, Martin Walter, Alkistis Skalkidou, 'Predicting women with depressive symptoms postpartum with machine learning methods', Article number: 7877 (2021)
- [5] Yiye Zhang, Shuojia Wang, Alison Hermann, Rochelle Joly, Jyotishman Pathak, 'Development and validation of a machine learning algorithm for predicting the risk of postpartum depression among pregnant women', Volume 279, 15 January 2021, Pages 1-8
- [6] Weina Zhang¹ Author Orcid Image, Han Liu² Author Orcid Image, Vincent Michael Bernard Silenzio³ Author Orcid Image, Peiyuan Qiu⁴ Author Orcid Image, Wenjie Gong¹ Author Orcid Image, 'Machine Learning Models for the Prediction of Postpartum Depression: Application and Comparison Based on a Cohort Study', Published on 30.4.2020 in Vol 8, No 4 (2020): April Article Authors
- [7] Paolo Cellini, Alessandro Piloni, Giuseppe Delvecchio c, Chiara Moltrasio, Paolo Brambilla, 'Machine learning in the prediction of postpartum depression: A review', Volume 309, 15 July 2022, Pages 350-357
- [8] Sam Andersson¹, Deepti R. Bathula², Stavros I. Iliadis¹, Martin Walter, Alkistis Skalkidou¹, 'Predicting women with depressive symptoms postpartum with machine learning methods'
- [9] HARINI MAHADEVAN, 'post partum depression analysis'